

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech/Int. B.Tech, Winter 2025-26 Examination

Semester: VII / XI

Subject Code: 303105341

Subject Name: Cyber Security

Date: 06-12-2025

Time: 10.30 am to 1.00 pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define the role of firewalls in cybersecurity. b) Explain how firewalls prevent unauthorized access. c) Mention two benefits of firewalls in protecting sensitive data.	06	CO1, CO5	BT3
	B. a) What is a vulnerability probe in network security? b) Mention two common vulnerabilities detected by vulnerability probes. c) Explain how probing helps in identifying weak network configurations.	06	CO1	BT2
Q.2	A. How does a firewall process works define in steps?	04	CO1	BT4
	B. How does it help access internal services remotely? Explain briefly.	05	CO5	BT1
	OR			
	B. Explain the significance of user involvement in system development.	05	CO3, CO4, CO6	BT1
Q.3	A. Explain the significance of which of the following is an example of a volumetric ddos attack? in a practical scenario.	04	CO3	BT1
	B. What does phishing mean? Identify three distinct phishing methods.	05	CO2	BT4
	OR			
	B. Why is what are common methods of password cracking? important in the field of cybersecurity?	05	CO3, CO4, CO6	BT2

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define cyberbullying and conventional bullying. b) Mention two key differences between cyberbullying and conventional bullying. c) Explain how the anonymity of the internet impacts cyberbullying.	06	CO1, CO5	BT2
	B. a) Why are information systems important in modern organizations? b) How do information systems improve decision-making in organizations? c) Mention two ways information systems enhance organizational efficiency.	06	CO1	BT1

Q.2	A. How can banner grabbing help identify outdated or insecure services on a network?	04	CO3	BT2
	B. What is the purpose of vulnerability scanning in network security?	05	CO2, CO3, CO4, CO6	BT3
	OR			
	B. Explain the role of Netcat in establishing network connections and detecting open ports.	05	CO3, CO4, CO6	BT5
Q.3	A. Can you outline how password cracking operates?	04	CO4	BT6
	B. What are the major signs of a security issue?	05	CO6	BT3
	OR			
	B. What are phishing attacks, and how can users protect themselves?	05	CO2, CO3, CO4, CO6	BT3